

Higher-Order Metaphysics

Week Six: How is HOL related to natural-language property-talk?

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1 Note on some abbreviations

2 Doing natural-language semantics in Higher-Orderese

Formal languages are good for the purposes of formulating precise, rigorous theories of many different subject matters. One subject matter where the resources of higher-order language are particularly useful is that of *semantics* in the sense of *theory of meaning*—an investigation that aims to identify and explain regularities within a population about the psychological states (e.g. beliefs) of those producing and observing different sounds / strings of symbols, regularities about the speech acts (e.g. assertion) accomplished by producing them, etc.¹

Semantics depends on theoretical terms. When it's done in English, there is a panoply of roughly-interchangeable ones—‘means’, ‘refers to’, ‘denotes’, ‘expresses’.... The natural first-pass way of rendering such talk in HOL is to have, for each type σ , we have a predicate E_σ of type $\kappa \rightarrow \varepsilon \rightarrow \sigma \rightarrow t$, where κ is the type corresponding to a name for the linguistic community we are trying to study ($e? e \rightarrow t?$) and ε is the type corresponding to quote-names for expressions of the language we are studying ($e? e \rightarrow t?$)

However, we are free to innovate in our choice of theoretical terms if doing so is useful for the purposes of formulating simple and explanatory theories.

One minor innovation is motivated by the fact that some words can play several different *syntactic* roles, and have different meanings depending on which role they play.²

To represent such relativity we can introduce new primitive semantic predicates $:_\sigma$ with an extra argument place for a *syntactic category*. For concreteness we can identify with a string (type ε)—e.g. ‘*n*’ for noun, ‘*s*’ for sentence, etc.³ So we can formulate hypotheses like th

$$\text{planet} : n \text{planet}^{\bar{e}} \\ \text{no-planet.twinkles} : s \rightarrow \exists x^e. \text{planet}^{\bar{e}} x \wedge \text{twinkles}^{\bar{e}} x$$

This leaves us free to cook up whatever notation for English syntactic categories best lends itself to the goal of formulating simple compos-

¹ I take “semantics” in this sense to be quite different from the enterprise of *model-theoretic semantics*, even though there is much overlap in practice. The model-theoretic semanticist might define a notion of ‘model for English’ in which models are set-theoretic constructs that include an arbitrary “domain”, and build it into the definition of ‘model’ that each model associates (nonoverlapping?) subsets of the domain with the English words ‘dog’ and ‘cat’. Such definitions may illuminate facts about which inferences ‘feel valid’ to English speakers, but don’t engage with the fact that there is a dog’ can be used to say that there is a dog and can’t (normally) be used to say that there is a cat.

² E.g., ‘study’ can function as a noun, an intransitive verb, or a transitive verb, and how it functions affects its semantic potential: ‘You study’ means something quite different from ‘You are a study’.

³ Henceforth I will designate this type as ε to avoid confusion with other uses of *et*.

I suppress the argument for a linguistic community, write the “expression” argument on the left, and suppress the subscript to $:_\sigma$ when it can be reconstructed from the type of the final argument.

itional laws. One nice approach⁴ builds up categories out of a small list of base categories using two directional slashes / and \. Expressions of categories c/d and d/c are ones that can combine with an expression of category d , occurring respectively on the right and on the left, to yield an expression of category c . We can regiment this in our higher-order language using two new constants:⁵

$$\begin{aligned} / : \varepsilon \rightarrow \varepsilon \rightarrow \varepsilon \\ \backslash : \varepsilon \rightarrow \varepsilon \rightarrow \varepsilon \end{aligned}$$

Using this system of syntactic categories, we can formulate two simple schemas that allow us to derive semantic claims about complex expressions from semantic claims about their simpler parts:

$$\begin{aligned} \text{Functional Applicaton (FA)} \quad (>) \quad a :_{\sigma \rightarrow \tau} c/d \ X \wedge b :_{\sigma} d \ y \rightarrow a \cdot b :_{\tau} c \ Xy \\ (<) \quad a :_{\sigma \rightarrow \tau} d/c \ X \wedge b :_{\sigma} d \ y \rightarrow b \cdot a :_{\tau} c \ Xy \end{aligned}$$

Example:

$$\begin{array}{c} \frac{\text{no} : (s/np/s)/n \quad \text{planet} : n}{\lambda X^{\bar{e}} Y^{\bar{e}}. \neg \exists z^e. Xz \wedge Yz \quad \text{planet}^{\bar{e}}} > \\ \frac{\text{no-planet} : s/np/s \quad (\lambda X^{\bar{e}} Y^{\bar{e}}. \neg \exists z^e. Xz \wedge Yz) \text{planet}^{\bar{e}}}{\text{no-planet} : s/np/s \quad \lambda Y^{\bar{e}}. \neg \exists z^e. \text{planet}^{\bar{e}} z \wedge Yz} \beta \\ \frac{\text{no-planet} : s/np/s \quad \lambda Y^{\bar{e}}. \neg \exists z^e. \text{planet}^{\bar{e}} z \wedge Yz \quad \text{twinkles} : np/s}{\text{no-planet} : s/np/s \quad \lambda Y^{\bar{e}}. \neg \exists z^e. \text{planet}^{\bar{e}} z \wedge Yz \quad \text{twinkles}^{\bar{e}}} > \\ \frac{\text{no-planet} : s/np/s \quad \lambda Y^{\bar{e}}. \neg \exists z^e. \text{planet}^{\bar{e}} z \wedge Yz \quad \text{twinkles}^{\bar{e}}}{\text{no-planet} : s/np/s \quad \lambda Y^{\bar{e}}. \neg \exists z^e. \text{planet}^{\bar{e}} z \wedge Yz \quad \text{twinkles}^{\bar{e}}} \beta \\ \text{no-planet-twinkles} : s \\ (\lambda Y^{\bar{e}}. \neg \exists z^e. \text{planet}^{\bar{e}} z \wedge Yz) \text{twinkles}^{\bar{e}} \\ \text{no-planet-twinkles} : s \\ \neg \exists z^e. \text{planet}^{\bar{e}} z \wedge \text{twinkles}^{\bar{e}} z \end{array}$$

- Derivations that look a lot like this have been a staple of semantics since Montague (1974a). But Montague conceived of them as characterizations of a *translation* from English to the logical language—for him, we are *mentioning* both the English expression and the formal expression that appear on each line. By contrast, here the *whole line* as a sentence of the formal language, in which the higher-order expression that is the last argument of $:_{\sigma}$ is *used*, not mentioned.

3 A note on the handling of ambiguity in this system

The setup allows in-principle possible forms of *ambiguity* (semantic multiplicity) for an expression a :

⁴ Bar-Hillel (1953), Lambek (1958)

⁵ To economize on parentheses we treat \ as binding more tightly than /, / as left-associative, and \ as right-associative: e.g., $a \backslash b / c$ is short for $(a \backslash b) / c$ and $a / b \backslash c$ is short for $a / (b \backslash c)$.

Gloss: when a has a type $\sigma \rightarrow \tau$ meaning X and b has a type- σ meaning y , and moreover the categories relative which they have these meanings are such as to license the combination, the result of concatenating a and b in the appropriate order has the type- τ meaning that results from applying X to y .

‘We could...introduce the semantics of our fragment directly; but it is probably more perspicuous to proceed indirectly, by (1) setting up a certain simple artificial language, that of tensed intensional logic, (2) giving the semantics of that language, and (3) interpreting English indirectly by showing in a rigorous way how to translate it into the artificial language.’ (Montague 1974b: 256).

For Montague, these translations are just prolegomena to “semantics”, which he equates with set-theoretic model theory.

- (i) a expresses distinct things of a certain type σ relative to distinct categories:⁶

$$\exists x^\sigma y^\sigma c^\epsilon d^\epsilon. a :_\sigma c x \wedge a :_\sigma d y \wedge x \neq_\sigma y \wedge c \neq_\epsilon d$$

- (ii) a expresses distinct things of a certain type σ relative to a single category:⁷

$$\exists x^\sigma y^\sigma c^\epsilon. a :_\sigma c x \wedge a :_\sigma c y \wedge x \neq_\sigma y$$

- (iii) a expresses things of distinct types σ and τ relative to distinct categories:⁸

$$\exists x^\sigma y^\tau c^\epsilon d^\epsilon. a :_\sigma c x \wedge a :_\tau d y$$

- (iv) a expresses things of distinct types σ and τ relative to a single category:

$$\exists x^\sigma y^\tau c^\epsilon. a :_\sigma c x \wedge a :_\tau c y$$

Following Montague (1974b), semantic theories using category-labels like ours have generally disallowed type-(iv) ambiguity—indeed even *distinct* expressions from having meanings of different types relative to the same category. This could be expressed by the following schema:

$$\text{Correspondence} \quad \neg(a :_\sigma c x \wedge b :_\tau c y) \quad \text{where } \sigma \neq \tau$$

If one really wanted this, one could always guarantee it by fine-graining the category-labels; however, nothing breaks if we reject it, and the theory to be explored below does reject it.

One further point about the treatment of ambiguity in this system is worth noting: each time we encounter an occurrence of an ambiguous word, we get to make a fresh choice from among all its meanings that are of an appropriate type and are had relative to an appropriate category:

$\text{some} : (s/np\backslash s)/n$ $\lambda X^e Y^e. \exists z^e. Xz \wedge Yz$	$\text{bank} : n$ $\mathbf{mbank}^e$	$\text{no} : (s/np\backslash s)/n$ $\lambda X^e Y^e. \neg \exists z^e. Xz \wedge Yz$	$\text{bank} : n$ $\mathbf{rbank}^e$	$>$
$\text{some} \cdot \text{bank} : s/np\backslash s$ $\lambda Y^e. \exists z^e. \mathbf{mbank}^e z \wedge Yz$		$\text{no} \cdot \text{bank} : s/np\backslash s$ $\lambda Y^e. \neg \exists z^e. \mathbf{rbank}^e z \wedge Yz$		$>$
$\text{lends} : np\backslash s$ $\mathbf{lender}^e$		$\text{and} : s\backslash s$ $\lambda p^t q^t. q \wedge p$		$>$
$\text{some} \cdot \text{bank} \cdot \text{lends} : s$ $\exists z^e. \mathbf{mbank}^e z \wedge \mathbf{lender}^e z$		$\text{no} \cdot \text{bank} \cdot \text{lends} : s$ $\neg \exists z^e. \mathbf{rbank}^e z \wedge \mathbf{lender}^e z$		$>$
$\text{and} \cdot \text{no} \cdot \text{bank} \cdot \text{lends} : s\backslash s$ $\lambda p^t. p \wedge \neg \exists z^e. \mathbf{rbank}^e z \wedge \mathbf{lender}^e z$		$\text{some} \cdot \text{bank} \cdot \text{lends} \cdot \text{and} \cdot \text{no} \cdot \text{bank} \cdot \text{lends} : s$ $(\exists z^e. \mathbf{mbank}^e z \wedge \mathbf{lender}^e z) \wedge \neg \exists z^e. \mathbf{rbank}^e z \wedge \mathbf{lender}^e z$		$>$

Often (though not always) speakers and hearers prefer *uniform* interpretations.⁹ In other work, I explore consider enriching the semantic ideology using with polyadic ‘co-expressing’ predicates, in order to give a theory of this notion of uniformity.

⁶ EG: the distinct type- $\bar{e}$ meanings of ‘study’ as noun (category n) and as verb phrase (category $np\backslash s$)

⁷ EG: the distinct meanings of ‘bat’ (as noun).

⁸ EG: the type- $\bar{e}$ meaning of ‘study’ as a verb phrase (category $np\backslash s$) and its type- $\bar{e}\bar{e}$ meaning as transitive verb (category $np\backslash s/np$).

⁹ The notion of uniformity is particularly important when we are interested in logical notions like consequence and contradictoriness (see Dorr 2014).

4 Higher-order logic and English ‘property’ talk

We have seen how *pronouncing* sentences of Higher-Orderese, it can be pedagogically helpful to associate them with English sentences containing words like ‘property’, ‘relation’, ‘proposition’, etc. For example, one might pronounce $\neg\forall X^e.X \text{mars} \rightarrow X \text{venus}$ as ‘Not every property of Mars is a property of Venus’, and $\exists p^t.\neg p$ as ‘Some proposition is not true’ or ‘Some state of affairs does not obtain’.

An interesting disputed question: do these stipulated mappings actually correspond to the literal meaning of those English sentences? Or do they (as, e.g., Bacon assumes) depart radically from the literal meanings? The question is naturally asked in Higher-Orderese with constants $:_\sigma$: is it the case, e.g., that

Some-proposition-is-not-true : $s \exists p^t.\neg p$

If our understanding of HOL depended entirely on our translating it into English along such lines, the answer to this question would be ‘obviously yes’. But if we have some other way of getting to understand HOL—e.g., learning it ‘by the direct method’ by mastering the rules of some system of logic—the question becomes an interesting one, both for semanticists interested in the workings of such expressions for their own sake, and for philosophers who care because they want to *use* the expressions to state philosophically interesting claims and arguments.

A thought experiment: we are alien field linguists (from the exoplanet Lambda Serpentis b), whose native language is Higher-Orderese.¹⁰ We are trying to figure out what English-speakers mean by various sentences.

¹⁰ With a rich signature Σ of non-logical constants.

So far this has been going well, thanks to the presence in our language of close analogues of many English words. For example, we have constants **planet** ^{$\bar{e}$} and **twinkles** ^{$\bar{e}$} , with which we can claims like the premises of the little derivation in the previous section. But we are puzzled when we try to extend our theory to English expressions like ‘property’, ‘relation’, ‘redness’, ‘instantiates’, etc.

Why would we be puzzled? Well, let’s suppose the following thesis is popular among philosophers on Lambda Serpentis:

Materialism_e $\forall x^e. \text{material } x$

‘**material**’ is a placeholder; we could cash it out in terms of having a shape, size, mass, parts, location....

It’s only natural that Materialism_e would be much more popular there than the following English sentence is on Earth:

Materialism Everything is a material object.

Philosophers on Earth who assert this sentence have a hard life, since

they immediately have to confront objections like the following, which strike many of their opponents as decisive:

Redness isn't a material object; so not everything is a material object.

The fact that there are dogs isn't a material object; so not everything is a material object.

The number two isn't a material object; so not everything is a material object.

The word 'cat' isn't a material object; so not everything is a material object.

By contrast, while proponents of Materialism_e on Lambda Serpents still have to reckon with arguments about gods, ghosts, souls, etc., they do not have to consider analogues of these objections, since their language lack type-*e* terms analogous to 'redness', 'the fact that there are dogs', or 'the number two'. Their closest analogues are terms of higher types.¹¹

Materialism_e makes the problem more vivid. 'Instantiates' is syntactically like 'loves' and 'redness' is syntactically like 'Mars'. Both are meaningful; so the natural hypothesis is that they are meaningful in the same types as these expressions:

$$(1) \quad \exists R^{\bar{ee}}. \text{instantiates} : np \backslash s / n \ R$$

$$(2) \quad \exists x^e. \text{redness} : np \ x$$

But what could this *R* and *x* be? Our materialist_e starting point doesn't suggest any good candidates, given how speakers treat 'Mars instantiates redness' as equivalent to 'Mars is red', and more generally treat 'Mars instantiates VP+ing' as equivalent to 'Mars VP'.

"Inflationary" option. Take English-speakers to be in the grip of an extravagant "Platonist" view, inconsistent with Materialism_e, which posits a realm of special individuals ("universals"), including a fit referent for 'redness'.¹²

*"Deflationary" option.*¹³ Reject the inference from syntactic uniformity to semantic non-uniformity. Instead, endorse the premises of derivations like this:

$$\frac{\text{instantiates} : np \backslash s / np \ \lambda X^{\bar{e}}. X \quad \text{redness} : np \ \text{red}^{\bar{e}}}{\text{instantiates} \cdot \text{redness} : np \backslash s \ ((\lambda X^{\bar{e}}. X) \ \text{red}^{\bar{e}})} >$$

$$\frac{\text{Mars} : np \ \text{mars}^e \quad \text{instantiates} \cdot \text{redness} : np \backslash s \ \text{red}^{\bar{e}}}{\text{Mars} \cdot \text{instantiates} \cdot \text{redness} : s \ (\text{red}^{\bar{e}} \ \text{mars}^e)} \beta$$

On this approach, it's natural to say that the difference between 'redness' and 'is red' is *merely* syntactic.¹⁴

5 Type-ambiguity

The 'deflationary' option requires pervasive type-ambiguity. For example:

¹¹ E.g. '2' is a term of type $et \rightarrow t$ equivalent to $\lambda X^{\bar{e}}. \exists y^e z^e. Xz \wedge Yz \wedge y \neq z$, and **number** is a term of type $(et \rightarrow t) \rightarrow t$.

The only analogues of quote-names are terms of type et : e.g., *catx* is true when *x* is a blob of ink shaped like this: cat.

¹² Button and Trueman (2024) develop an interesting "fictionalist" alternative, where the literal meanings require Platonism, but the sentences are not used literally.

¹³ See Parsons 1979.

¹⁴ Although someone might instead want to classify differences as regards the syntactic categories relative to which expressions express whatever they express as "semantic" differences. This seems a merely verbal issue.

- ‘some’ will have to not only express $\lambda X^{\bar{e}} \bar{Y}^{\bar{e}} . \exists z^e . Xz \wedge Yz$, but also $\lambda X^{\bar{e}} \bar{Y}^{\bar{e}} . \exists z^{\bar{e}} . (Xz \wedge Yz)$, to allow for derivations like this:

$$\begin{array}{c}
 \begin{array}{c}
 \text{of} : n \backslash n / np \\
 \lambda x^e Z^{e \rightarrow et \rightarrow t} . Zx \\
 \hline
 \text{property} : n \quad \text{of-Venus} : n \backslash n \\
 \lambda x^e \bar{Y}^{\bar{e}} . Yx \quad \lambda Z^{e \rightarrow et \rightarrow t} . Z \text{ venus} \\
 \hline
 \text{some} : s / np \backslash s / n \quad \text{property-of-Venus} : n \\
 \lambda X^{et \rightarrow t} \bar{Y}^{et \rightarrow t} . \exists X^{\bar{e}} . Xz \wedge Yz \quad \lambda Y^{\bar{e}} . Y \text{ venus} \\
 \hline
 \text{some-property-of-Venus} : s / np \backslash s \\
 \lambda Y^{\bar{e}} . \exists X^{\bar{e}} . X \text{ venus} \wedge YX
 \end{array}
 >
 \begin{array}{c}
 \text{Venus} : np \\
 \text{venus} \\
 \hline
 \text{of} : (np \backslash s) \backslash np \backslash s / np \quad \text{Mars} : np \\
 \lambda x^e Z^{e \rightarrow et \rightarrow t} . Zx \quad \text{mars} \\
 \hline
 \text{holds} : np \backslash s \quad \text{of-Mars} : (np \backslash s) \backslash np \backslash s \\
 \lambda x^e \bar{Y}^{\bar{e}} . Yx \quad \lambda Z^{e \rightarrow et \rightarrow t} . Z \text{ mars} \\
 \hline
 \text{holds-of-Mars} : np \backslash s \\
 \lambda Y^{\bar{e}} . Y \text{ mars}
 \end{array}
 <
 \end{array}
 >
 \begin{array}{c}
 \text{some-property-of-Venus-holds-of-Mars} : s \\
 \exists X^{\bar{e}} . X \text{ venus} \wedge X \text{ mars}
 \end{array}$$

- ‘interesting’ will express something of type et (‘Mars is interesting’); something of type $\bar{e}$ (‘Redness is interesting’); something of type $(et \rightarrow t) \rightarrow t$ (‘Being instantiated by Mars is interesting’)...¹⁵
- ‘mentions’ will, likewise, have to express something of type $\bar{e}\bar{e}$; something of type $et \rightarrow et$; something of type $(et \rightarrow t) \rightarrow et$...

¹⁵ Our field linguists might perfectly well have appropriate constants **interesting_e**, **interesting_e**, **interesting_{et \rightarrow t}**...

In fact, type-ambiguity will need to be quite pervasive, if we don’t want to have to count some OK-seeming sentences are meaningless!¹⁶ E.g., I think that

(3) Redness is red

had better be meaningful, since it occurs as part of clearly meaningful sentences like ‘Few philosophers would claim that redness is red’. Since ‘redness’ has no type- e meaning, this means we’ll need

- To do so, we’ll need a type- $\bar{e}$ meaning M for ‘is red’: plausibly, one for which $\Box \forall \bar{e} X \neg MX$ is true, perhaps necessarily true. One simple option is to say $\text{is-red} : np \backslash s \lambda X^{\bar{e}} . \perp$, so that $\text{redness-is-red} : s \perp$.
- Similarly, ‘instantiates’ will have to not only express $\lambda X^{\bar{e}} . X$, $\lambda X^{\bar{e}} . X$, etc., but also some R of type et , to explain the meaningfulness of ‘Mars instantiates Venus’ (and the truth of its negation). Plausibly this R is such that $\forall x^e y^e . \neg Rxy$. Again, a simple option is to say that $E_{\bar{e}\bar{e}}$ instantiates $\lambda x^e y^e . \perp$.
- General conclusion: every meaningful English verb phrase and common noun has a meaning in every type $\bar{o}$. And every transitive verb has a meaning in every type $\bar{o}_1 \bar{o}_2$.¹⁷

¹⁶ Prior did just this: ‘For “Percy is a fact” (which would mean “It is the case that Percy”, if it meant anything), “Percy is a falsehood” (=“It is not the case that Percy”), ‘Percy is neither a fact nor a falsehood (=“It neither is nor is not the case that Percy”) are all of them senseless, ungrammatical.’ (Prior 1971: 25)

Alternatively, we could say that *word* is enormously *vague* in type $\bar{e}$.

¹⁷ Schematically::

$$(\exists x^{\bar{o}_1} . a :_{\bar{o}_1} n x) \rightarrow (\exists y^{\bar{o}_2} . a :_{\bar{o}_2} n x)$$

and similarly for the other categories.

6 Type-ambiguity and Russell's paradox

Black-boxing the syntactic operation involved in forming gerunds like 'being red', it's natural to propose the following law:

Gerund $\forall a^e X^{\bar{\sigma}}.a :_{\bar{\sigma}} np \backslash s X \rightarrow a + ing :_{\bar{\sigma}} np X$

Given this, every instance of NPC has a true reading.

Naïve Property Comprehension [NP] [VP] if and only if [NP] instantiates [VP+ing].

But of course they won't *exclusively* have true readings. Our theory is not a theory of "uniform meaning": when a sentence has multiple occurrences of an ambiguous expression, we can interpret them differently, e.g. giving a false reading for

(4) Barclays is a bank if and only if Barclays instantiates being a bank

and likewise (assuming $\mathbf{int}^{\bar{e}} \mathbf{red}^{\bar{e}}$) for

(5) Being red is interesting if and only if being red instantiates being interesting.

Non-uniform readings are usually less salient. But for some NPC-instances, *all* readings involve non-uniformity:

(6) Being interesting is interesting if and only if being interesting instantiates being interesting.

Likewise for:

(7) Not instantiating oneself does not instantiate itself if and only if not instantiating oneself instantiates not instantiating oneself.

Following Szabolsci (1987), we can treat 'itself' as a reflexiviser:

$\text{itself} : (np \backslash s / np) \backslash (np \backslash s) \lambda X^{\bar{\sigma}\bar{\sigma}} y^{\sigma}.Xyy$

Given our posits, it follows that $\forall X^{\bar{\sigma}}.\text{instantiates} \cdot \text{itself} : np \backslash s R \rightarrow \forall y^{\sigma}.\neg Xy$, so everything expressed by the LHS of (7) is true. By contrast, the RHS admits both false readings (where the two occurrences of 'not instantiating oneself' are interpreted the same type $\bar{\sigma}$) and true ones (where the first is $\bar{\sigma}$ and the second is $\bar{\bar{\sigma}}$).

The RHS has a false reading where 'being red' expresses $\mathbf{red}^{\bar{e}}$, 'being interesting' expresses $\mathbf{int}^{\bar{e}}$, and 'instantiates' expresses something of type $\bar{e}\bar{e}$: recall that we take 'instantiates' to express empty relations in all types $\bar{\sigma}_1\bar{\sigma}_2$ where σ_1 is not $\bar{\sigma}_2$.

In the necessarily-true readings, the first and third 'interesting' express $\mathbf{int}^{\bar{\sigma}}$, while the second and fourth express $\mathbf{int}^{\bar{\bar{\sigma}}}$, for some σ . But another salient option (which makes the RHS false) is to take the first, third, and fourth to express $\mathbf{int}^{\bar{\sigma}}$ and the second to express $\mathbf{int}^{\bar{\bar{\sigma}}}$.

$$\begin{array}{l} \text{loves} : np \backslash s / np \quad \text{itself} : (np \backslash s / np) \backslash np \backslash s \\ \text{loves}^{\bar{e}\bar{e}} \quad \lambda X^{\bar{e}\bar{e}} y^e.Xyy \\ \hline \text{loves} \cdot \text{itself} : np \backslash s \\ \lambda y. \text{loves}^{\bar{e}\bar{e}} yy \end{array} <$$

7 The co-ordination problem

One standard objection¹⁸ to a higher-order treatment of property-talk in natural language involves sentences like

(8) [Either] Mars or redness is interesting.¹⁹

We'd like this to turn out to be interchangeable with

(9) Mars is interesting or redness is interesting.

We can explain why (8) expresses **interesting_e mars** $\vee$ **interesting_e red**, by invoking two differently-typed meanings for 'interesting'. But how can we arrange this for (8), where we only have one occurrence to work with?²⁰

I am going to tackle this by giving a semantic theory for English *stated in a higher-order language with sum-types*. It looks very much the same as the theory we've been working with now, except that it has constants $\cdot_\alpha$ for extended types as well as simple types: e.g., $\cdot_{e+\bar{e}}$, $\cdot_{\bar{e}+\bar{e}}$, etc. It still has the function-application schemas FA (now understood to encompass the new types). And it has one new general law that lets us type-shift arbitrary expressions to sum-types:

$$\begin{aligned} \text{Inject} \quad & a :_\alpha c x \rightarrow a :_{\alpha+\beta} c \iota_\beta^1 x \\ & a :_\beta c y \rightarrow a :_{\alpha+\beta} c \iota_\alpha^2 y \end{aligned}$$

Switching to this language isn't a big deal, since the new constants are all of types isomorphic to simple types, and thus there is a systematic translation from that theory to a theory (with new constants) given in the original simple Higher-Orderese. For example, instead of the simple constant $\cdot_{e+\bar{e}}$ (of type $\varepsilon\varepsilon\bar{e} + \bar{e}$), we will have a pair of constants $\cdot_{e+\bar{e}}'$ of type $\varepsilon\varepsilon\bar{e}$ and $\cdot_{e+\bar{e}}''$ of type $\varepsilon\bar{e}\bar{e}$, and instead of $\cdot_{\bar{e}+\bar{e}}$ (of type $\varepsilon\bar{e}\bar{e} + \bar{e}$), we'll have a constant $\cdot_{\bar{e}+\bar{e}}'$ of type $\varepsilon\bar{e}\bar{e}$.

8 Preliminaries to a solution 1: 'or' and type-lifting

The other thing we'll need is an account of the co-ordinating conjunction 'or' (and 'and') that can handle non-mixed cases like (10) and (11):

(10) [Either] Mars or Venus is interesting.

(11) [Either] redness or blueness is interesting.

For concreteness, I'll choose the best-known approach,²¹ This turns on

¹⁸ Bealer 1993, Hofweber 2022, Button and Trueman 2024

¹⁹ We could make the same point using 'and'; 'or' is just slightly easier because it doesn't require plural marking.

²⁰ 'When I was young, the things I cared most about were things that I could see or feel, but now they are things I can know to be true'. Etc. Bealer 1993: 9, n. 8

²¹ Partee and Rooth (1983), generalizing the earlier work of Montague (1974b).

the following type-shifting law:

$$\begin{aligned} \text{Lift} \quad & a :_{\sigma} c x \rightarrow a :_{(\sigma \rightarrow \tau) \rightarrow \tau} (d/c) \backslash d \lambda Y^{\sigma \rightarrow \tau}. Yx \\ & a :_{\sigma} c x \rightarrow a :_{(\sigma \rightarrow \tau) \rightarrow \tau} d/(c \backslash d) \lambda Y^{\sigma \rightarrow \tau}. Yx \end{aligned}$$

We can combine this with a type-ambiguous treatment of *or* where it expresses disjunction in every type ending in *t*. This can be captured with the following postulates:

$$\begin{aligned} (12) \quad & (a) \quad \text{or} : s \backslash s / s (\vee) \\ & (b) \quad \text{or} : n \backslash n / n \lambda X^{\sigma} Y^{\sigma} Z^{\sigma}. Xz \vee Yz \\ & (c) \quad \forall c^{\varepsilon} d^{\varepsilon} R^{\tau \rightarrow \tau \rightarrow \tau}. \text{or} : c \backslash c / c R \rightarrow \text{or} : (c/d) \backslash (c/d) / (c/d) \lambda X^{\sigma \rightarrow \tau} Y^{\sigma \rightarrow \tau} Z^{\sigma}. R(Xz)(Yz) \\ & (d) \quad \forall c^{\varepsilon} d^{\varepsilon} R^{\tau \rightarrow \tau \rightarrow \tau}. \text{or} : c \backslash c / c R \rightarrow \text{or} : (c \backslash d) \backslash (c \backslash d) / (c \backslash d) \lambda X^{\sigma \rightarrow \tau} Y^{\sigma \rightarrow \tau} Z^{\sigma}. R(Xz)(Yz) \end{aligned}$$

Here is an account of (11) using these ingredients:

$$\begin{array}{c} \frac{\text{redness} : np \text{red}^{\bar{e}}}{\text{redness} : s/np \backslash s} \text{Lift} \quad \frac{\frac{\text{or} : s \backslash s / s (\vee) \quad 18a}{\text{or} : (s/np \backslash s) \backslash (s/np \backslash s) / (s/np \backslash s)} \quad 18c \quad \frac{\text{blueness} : np \text{blue}^{\bar{e}}}{\text{blueness} : s/np \backslash s} \text{Lift}}{\lambda U^{(et \rightarrow t) \rightarrow t} V^{(et \rightarrow t) \rightarrow t} X^{\bar{e}}. VX \vee UX} \quad \frac{\lambda X^{\bar{e}}. X \text{blue}^{\bar{e}}}{\text{or-blueness} : (s/np \backslash s) \backslash (s/np \backslash s) \lambda V^{(et \rightarrow t) \rightarrow t} X^{\bar{e}}. VX \vee X \text{blue}^{\bar{e}}} > \\ \frac{\text{redness-or-blueness} : s/np \backslash s \lambda X^{\bar{e}}. X \text{red}^{\bar{e}} \vee X \text{blue}^{\bar{e}} \quad \text{is-interesting} : np \backslash s \text{int}^{\bar{e}}}{\text{redness-or-blueness-is-interesting} : s \text{int}^{\bar{e}} \text{red}^{\bar{e}} \vee \text{int}^{\bar{e}} \text{blue}^{\bar{e}}} < > \end{array}$$

The treatment of (10) will be parallel, substituting *e* for *et* throughout.

9 Explaining mixed co-ordination

We just need one other thing we need to explain the puzzling mixed predications, namely an extension of our type-ambiguous treatment of predicates like ‘is interesting’ that also provides meanings in extended types like $e + et \rightarrow t$. I’ll assume that when the argument is a sum-type, this is just the result of combining the meanings in the summands:

$$(13) \quad \text{is-interesting} : np \backslash s \langle \langle \text{int}^{\bar{e}}, \text{int}^{\bar{e}} \rangle \rangle$$

Here, $\langle \langle F, G \rangle \rangle$ is short for $\lambda x^{e+\bar{e}}. \delta x(u. Fu)(v. Gv)$.²²

With these assumptions, we can derive the meaning we want for (8) as follows.²³

²² Thus given the conversion rules of H for the extended language, $\langle \langle F, G \rangle \rangle_{t_{\beta}}^1 A$ converts to FA and $\langle \langle F, G \rangle \rangle_{t_{\alpha}}^2 B$ converts to GB .

²³ (We abbreviate the type $e + et \rightarrow t$ as J and the category $s/np \backslash s$ as $np^{\dagger}$).

$$\begin{array}{c}
\text{redness : } np \\
\text{red}^{\bar{e}} \\
\hline
\text{redness : } np \\
l_e^2 \text{red}^{\bar{e}} \\
\hline
\text{Inject}
\end{array}
\quad
\begin{array}{c}
\text{redness : } np \\
l_e^2 \text{red}^{\bar{e}} \\
\hline
\text{Lift}
\end{array}$$

$$\begin{array}{c}
\text{Mars : } np \\
\text{mars}^e \\
\hline
\text{Inject}
\end{array}
\quad
\begin{array}{c}
\text{or : } np^{\uparrow} \backslash np^{\uparrow} / np^{\uparrow} \\
\lambda U^{\bar{e}+\bar{e}} V^{\bar{e}+\bar{e}} X^{\bar{e}+\bar{e}} . VX \vee UX \\
\hline
\text{Lift}
\end{array}$$

$$\begin{array}{c}
\text{Mars : } np^{\uparrow} \\
\lambda X^{\bar{e}+\bar{e}} . X l_e^1 \text{mars}^e \\
\hline
\text{Lift}
\end{array}
\quad
\begin{array}{c}
\text{or-redness : } np^{\uparrow} \backslash np^{\uparrow} \\
\lambda V^{\bar{e}+\bar{e}} X^{\bar{e}+\bar{e}} . VX \vee X l_e^2 \text{red}^{\bar{e}} \\
\hline
>
\end{array}$$

$$\begin{array}{c}
\text{Mars-or-redness : } np^{\uparrow} \\
\lambda X^{\bar{e}+\bar{e}} . X l_e^1 \text{mars}^e \vee X l_e^2 \text{red}^{\bar{e}} \\
\hline
<
\end{array}
\quad
\begin{array}{c}
\text{is-interesting : } np \backslash s \\
\langle\langle \text{int}^{\bar{e}}, \text{int}^{\bar{e}} \rangle\rangle \\
\hline
>
\end{array}$$

$$\begin{array}{c}
\text{Mars-or-redness-is-interesting : } s \\
\langle\langle \text{int}^{\bar{e}}, \text{int}^{\bar{e}} \rangle\rangle l_e^1 \text{mars}^e \vee \langle\langle \text{int}^{\bar{e}}, \text{int}^{\bar{e}} \rangle\rangle l_e^2 \text{red}^{\bar{e}} \\
\hline
\text{conv}
\end{array}$$

$$\begin{array}{c}
\text{Mars-or-redness-is-interesting : } s \\
\text{int}^{\bar{e}} \text{mars}^e \vee \text{int}^{\bar{e}} \text{red}^{\bar{e}}
\end{array}$$

We might be tempted to derive (13) from a general schema:

Combination $a :_{\alpha \rightarrow \gamma} c \ x \wedge a :_{\beta \rightarrow \gamma} c \ y \rightarrow a :_{\alpha + \beta \rightarrow \gamma} c \ \langle\langle x, y \rangle\rangle$

But this will generate undesirable results in other cases, e.g.

Barclays-or-Willow-is-a-bank : $s \text{riverbank barclays} \vee \text{moneybank willow}$

Instead, we should see ?? as capturing a distinctive kind of harmony between the type $\bar{e}$ and type $\bar{e}$ meanings we have posited for *is interesting*.

10 Sum-types and cross-type quantification

Sum-types also help with another well-known objection to the type-ambiguity approach. Plausibly,

(14) Everything Mary mentioned is interesting

has a reading where it both entails ‘If Mary mentioned Mars, Mars is interesting’ and ‘If Mary mentioned redness, redness is interesting’. We can get this by extending the type-ambiguity of the quantifiers to cover the sum-types, e.g.:

$$\begin{array}{l}
\text{every : } (s/np \backslash s)/n \ \lambda X^{\bar{e}+\bar{e}} Y^{\bar{e}+\bar{e}} . \forall z^{\bar{e}+\bar{e}} . Xz \rightarrow Yz \\
\text{everything : } s/np \backslash s \ \forall_{\bar{e}+\bar{e}}
\end{array}$$

In the version of Classicism for the extended language, we can prove the following equation

$$(15) \quad \forall_{\bar{e}+\bar{e}} =_{\bar{e}+\bar{e}} \lambda X^{\bar{e}+\bar{e}} . (\forall y^{\bar{e}} . X l_e^1 y) \wedge (\forall z^{\bar{e}} . X l_e^2 z)$$

Given this equation, we get the effect of quantifying in several types at once:

$$\begin{array}{c}
\text{everything} : s/np \backslash s \quad \text{is-interesting} : np \backslash s \\
(\lambda X^{e+\bar{e}}.(\forall y^e.X t_e^1 y) \wedge (\forall z^{\bar{e}}.X t_e^2 z)) \quad \langle\langle \text{int}^{\bar{e}}, \text{int}^{\bar{e}} \rangle\rangle \\
\hline
\text{everything-is-interesting} : s \\
(\forall y^e.\langle\langle \text{int}^{\bar{e}}, \text{int}^{\bar{e}} \rangle\rangle t_e^1 y) \wedge (\forall z^{\bar{e}}.\langle\langle \text{int}^{\bar{e}}, \text{int}^{\bar{e}} \rangle\rangle t_e^2 z) \\
\hline
\text{everything-is-interesting} : s \\
(\forall y^e.\text{int}^{\bar{e}} y) \wedge (\forall z^{\bar{e}}.\text{int}^{\bar{e}} z)
\end{array}
\quad \text{conv}$$

Even if we don't like the full strength of Classicism, the equation (15) looks metaphysically plausible in the context of H. Or we could just posit that 'everything' expresses the thing on the right, leaving it open whether this is the same as the thing on the left.²⁴

This solution isn't limited to quantifiers like 'every' and 'some' that commute with conjunction or disjunction. For example, 'Most Fs are G' can plausibly be analysed as 'whenever every *F*-and-*G* thing bears a relation to some *F*-and-not-*G* thing, there are two things that bear that relation to the same thing', i.e. $\text{most} : (s/np \backslash s)/n \text{ most}^{at \rightarrow at \rightarrow t}$, where

$$\text{most}^{\bar{\alpha}\bar{\alpha}} := \lambda F^{\bar{\alpha}} G^{\bar{\alpha}}. \forall R^{\bar{\alpha}\bar{\alpha}}. (\forall x^{\alpha}. (Fx \wedge Gx) \rightarrow (\exists y^{\alpha}. Fy \wedge \neg Gy \wedge Rxy)) \rightarrow (\exists x^{\alpha} y^{\alpha} z^{\alpha}. Rxz \wedge Ryz \wedge y \neq_{\alpha} z)$$

This gives plausible cross-type meanings for sentences like 'Most essay topics are interesting'.

Note that the cross-type entries for quantifier words provide true interpretations for some characteristic utterances of anti-nominalists, like 'Some but not all things are material, since Mars is material and redness is not'. We can still plausibly interpret many *nominalists* (e.g. Dorr 2007) who say 'Everything is material' to be speaking truly using a first-order quantifier.²⁵

²⁴ In the paper, I don't even let primitive quantifiers for extended types into the language in the first place.

²⁵ Like other kinds of ambiguity, type-ambiguity generates possibilities for merely verbal disagreements.

11 Worries about schemas and universal generalizations

A striking fact about the English sentence 'Everything is self-identical' is that everything it expresses is true, i.e.:

$$(16) \quad \forall p^t. (\text{everything-is-self-identical} : s p) \rightarrow p$$

This cries out for an explanation in terms of some facts about the constituents 'everything' and 'is self-identical'! But if both of them are type-ambiguous, it is hard to see how we could achieve this. We could certainly endorse schemas like the following: This cries out for an explanation in terms of some facts that the sentence in question is the result of combining 'everything' and 'is self-identical', together with some facts about those constituents. But given that both of them are type-ambiguous, it is hard to see how we could achieve this. We could certainly endorse schemas like the following: This cries out for an explanation

ation in terms of some facts that the sentence in question is the result of combining ‘everything’ and ‘is self-identical’, together with some facts about those constituents. But given that both of them are type-ambiguous, it is hard to see how we could achieve this. We could certainly endorse schemas like the following:

$$(17) \quad \forall X^{\bar{\sigma}}. \text{everything} : s/np \backslash s X \rightarrow X =_{\bar{\sigma}} \forall_{\sigma}$$

$$(18) \quad \forall Y^{\bar{\sigma}}. \text{is-self-identical} : np \backslash s Y \rightarrow Y =_{\bar{\sigma}} \lambda z^{\sigma}. z =_{\sigma} z$$

But we cannot derive (16) from the collection of all instances of these schemas—obviously, since we obviously can’t derive it from any finite subset.

A theorist who rejected type-ambiguity could try explaining (16) using invoke “unique decomposition” principles which limit the interpretations of sentences of the relevant form to those that can be derived by FA in the types dictated by their syntactic categories:²⁶

$$(19) \quad \forall a^{\epsilon} b^{\epsilon} p^{\dagger}. a \cdot b :_i s p \wedge \mathbf{Syn} a s/np \backslash s \wedge \mathbf{Syn} b np \backslash s \\ \rightarrow \exists X^{\bar{e}} Y^{\bar{e}}. a :_{\bar{e}} np \backslash s X \wedge b :_{\bar{e}} np \backslash s Y \wedge p = XY$$

In conjunction with the $\sigma = e$ instances of (17) and (18) and the unproblematic syntactic claims **Syn** everything $s/np \backslash s$ and **Syn** is-self-identical $np \backslash s$, (19) implies (16). Unfortunately, this strategy depends crucially on rejecting systematic type-ambiguity.

What can the lover of type-ambiguity do? I suggest adding *disquotational meaning predicates* M_{σ} for the original higher-order language (i.e. the one without the new constants).²⁷ We’ll need a background theory of these that includes **Reflection** as well as all instances of **DM**:²⁸

$$\mathbf{DM} \quad \forall x^{\sigma}. \ulcorner A \urcorner M_{\sigma} x \leftrightarrow x =_{\sigma} A \quad \text{where } A \text{ is closed and lacks any } M_{\sigma}$$

$$\mathbf{Reflection} \quad \forall a^{\epsilon}. \mathbf{TheoremOfH} a \rightarrow \mathbf{T} a$$

This seems legitimate even though by Tarski’s theorem on the undefinability of truth, there is no way these new predicates M_{σ} can be *defined* in the old language.

Using the disquotational resources, we can strengthen our semantic theory for English in a way that allows it to be finitely axiomatized (thus eliminating the need to rely on schemas). E.g. For example, instead of the schemas (17) and (18), we can write down single universally quantified sentences that (given the background theory) imply all their instances:²⁹

$$(20) \quad \forall a^{\epsilon}. \mathbf{Typ} a \rightarrow \mathbf{T} \ulcorner \forall X^{\bar{\$a}}. \text{everything} : np \backslash s X \rightarrow X =_{\bar{\$a}} \forall_{\$a} \urcorner$$

$$(21) \quad \forall a^{\epsilon}. \mathbf{Typ} a \rightarrow \mathbf{T} \ulcorner \forall Y^{\bar{\$a}}. \text{is-self-identical} : np \backslash s Y \rightarrow Y =_{\bar{\$a}} (\lambda z^{\$a}. z =_{\$a} z) \urcorner$$

²⁶ **Syn** : $\bar{\epsilon\epsilon}$ is a constant meaning ‘is the syntactic category of’.

²⁷ In terms of M_t we define “disquotational truth” predicate **T** is defined as $\lambda a^{\epsilon}. (\exists p^{\dagger}. a M_t p) \wedge (\forall p^{\dagger}. a M_t p \rightarrow p)$.

²⁸ $\ulcorner A \urcorner$ stands for a character-by-character specification of the string A . **TheoremOfH** abbreviates a syntactic definition that is tedious but straightforward given the Fregean higher-order definition of the ancestral.

²⁹ Here, the convention for corner-quotes is extended to allow for “anti-quotation”: a type- ϵ variable preceded by $\$$ still functions as a variable, analogous to the convention for Greek-lettered variables in Quine’s corner-quotes.

Typ is a predicate characterizing those strings that are well-formed types in our type system.

Since disquotational-meaning predicates give an analogue of *existential* as well as universal quantification into type position, we can formulate partial converses of FA such as:³⁰

$$(22) \quad \forall a^\epsilon b^\epsilon c^\epsilon p^t. \mathbf{Syn} a s/c \wedge \mathbf{Syn} b c \wedge a \cdot b :_t s p \rightarrow \exists f^\epsilon Z^t. \mathbf{Typ} c \wedge \\ \ulcorner \lambda q^t. \exists X^{\$f \rightarrow t} y^{\$f}. \$a :_{\$f \rightarrow t} s/c X \wedge \$b :_{\$f} c y \wedge q =_t X y \urcorner M_{\bar{t}} Z \wedge Z p$$

Thanks to the presence of **Reflection** in the background theory, (20), (21), and (23) will imply (16), via

$$(23) \quad \forall p^t. \text{everything-is-self-identical} : s p \rightarrow \\ \exists f^\epsilon Z^t. \mathbf{Typ} f \wedge \ulcorner \lambda q^t. q = \forall z^{\$f}. z =_{\$f} z \urcorner M_{\bar{t}} Z \wedge Z p$$

³⁰ (Gloss: for any sentence formed by concatenating a *s/c* expression *a* with a *c b* and any proposition *p* it expresses, there is a type *f* such that *p* has the property *Z* disquotationally-expressed by ‘is a *q* such that *q* is the result of applying something expressed by *a* in type *f* to something expressed by *b* in type *f*’.)

12 Type-neutral generality in English?

For any finitely many simple types $\sigma_1, \dots, \sigma_n$, we can construct a sum-type to generate interpretations for (24) and (25) equivalent respectively to (26) and (27):

(24) Every essay topic is interesting.

(25) Some essay topic is interesting.

$$(26) \quad (\forall x^{\sigma_1}. \mathbf{topic}^{\bar{\sigma}_1} x \rightarrow \mathbf{int}^{\bar{\sigma}_1} x) \wedge \dots \wedge (\forall x^{\sigma_n}. \mathbf{topic}^{\bar{\sigma}_n} x \rightarrow \mathbf{int}^{\bar{\sigma}_n} x)$$

$$(27) \quad (\exists x^{\sigma_1}. \mathbf{topic}^{\bar{\sigma}_1} x \wedge \mathbf{int}^{\bar{\sigma}_1} x) \vee \dots \vee (\exists x^{\sigma_n}. \mathbf{topic}^{\bar{\sigma}_n} x \wedge \mathbf{int}^{\bar{\sigma}_n} x)$$

But it’s tempting to think that (24) can also express something than all these, and (25) something weaker.

In Higher-Orderese with disquotational meaning predicates, such meanings could be expressed as follows:

$$(28) \quad \forall f^\epsilon. \mathbf{Typ} f \rightarrow \mathbf{T} \ulcorner \forall x^{\$f}. \mathbf{topic}^{\$f \rightarrow t} x \rightarrow \mathbf{int}^{\$f \rightarrow t} x \urcorner$$

$$(29) \quad \exists f^\epsilon. \mathbf{Typ} f \wedge \mathbf{T} \ulcorner \exists x^{\$f}. \mathbf{topic}^{\$f \rightarrow t} x \wedge \mathbf{int}^{\$f \rightarrow t} x \urcorner$$

So it’s not silly to wonder whether English affords a similar expressive possibility.

A possible strategy: postulate that *if a sentence expresses the conjunction/disjunction of any finite subcollection of some collection of propositions, it also expresses the conjunction/disjunction of all of them.*

To say this sort of thing in HOL, we can introduce a system of predicates $\mathbf{R}_\sigma$ of type $\bar{\sigma}$ expressing a notion of *rigidity*, where $\mathbf{R}_\sigma X^{\bar{\sigma}}$ implies that *X* is modally rigid in the way that sets are considered to be. Then we can understand quantification over ‘collections’ of propositions as

quantification over rigid properties of type $\bar{t}$, and equate the conjunction [disjunction] of a collection of propositions with the proposition that all [some] of them are true, and similarly for properties and relations:

$$\begin{aligned}\mathbf{Conj} \overline{X^{\sigma_1 \dots \sigma_n}} &:= \lambda y_1^{\sigma_1} \dots y_n^{\sigma_n}. \forall Z^{\overline{\sigma_1 \dots \sigma_n}}. XZ \rightarrow Zy_1 \dots y_n \\ \mathbf{Disj} \overline{X^{\sigma_1 \dots \sigma_n}} &:= \lambda y_1^{\sigma_1} \dots y_n^{\sigma_n}. \exists Z^{\overline{\sigma_1 \dots \sigma_n}}. XZ \wedge Zy_1 \dots y_n\end{aligned}$$

In these terms, the proposed new rules can be expressed by the following schemas:³¹

Conjunctive Closure

$$\forall a^{\epsilon} c^{\epsilon} X^{\bar{\tau}}. \mathbf{R}_{\tau} X \wedge (\forall Y^{\bar{\tau}}. \mathbf{R}_{\tau} Y \wedge \mathbf{Finite} Y \wedge Y \subseteq X \rightarrow a :_{\tau} c \mathbf{Conj} Y) \rightarrow a :_{\tau} c \mathbf{Conj} X$$

Disjunctive Closure

$$\forall a^{\epsilon} c^{\epsilon} X^{\bar{\tau}}. \mathbf{R}_{\tau} X \wedge (\forall Y^{\bar{\tau}}. \mathbf{R}_{\tau} Y \wedge \mathbf{Finite} Y \wedge Y \subseteq X \rightarrow a :_{\tau} c \mathbf{Disj} Y) \rightarrow a :_{\tau} c \mathbf{Disj} X$$

³¹ τ is any type ending in t . **Finite** is short for a standard higher-order definition of finitude, and $Y \subseteq X$ just means $\forall z^{\tau}. Yz \rightarrow Xz$.

However, adding the closure principles will inevitably disrupt certain logical relationships:

(30) P1 Some essay topic instantiates the property of being interesting.

P2 Jim mentioned the property of being interesting.

C Some essay topic instantiates some property Jim mentioned.

Assuming we posit a meaning for ‘mentioned’ of type $\bar{e}\bar{\alpha}$ for every extended type α , our old sum-type-based treatment of quantifiers is compatible with (30) having following nice status: for every proposition p expressed by P1, there are corresponding propositions q and r expressed by P2 and C respectively, such that $\Box((p \wedge q) \rightarrow r)$. But if we add the closure principles, it is unclear why (30) would have this status. The problem is that while P1 and C will both get maximally weak readings that behave like the disjunction of their old readings, P2 will not get any new readings, since its old readings were not closed under finite conjunction or disjunction.

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